

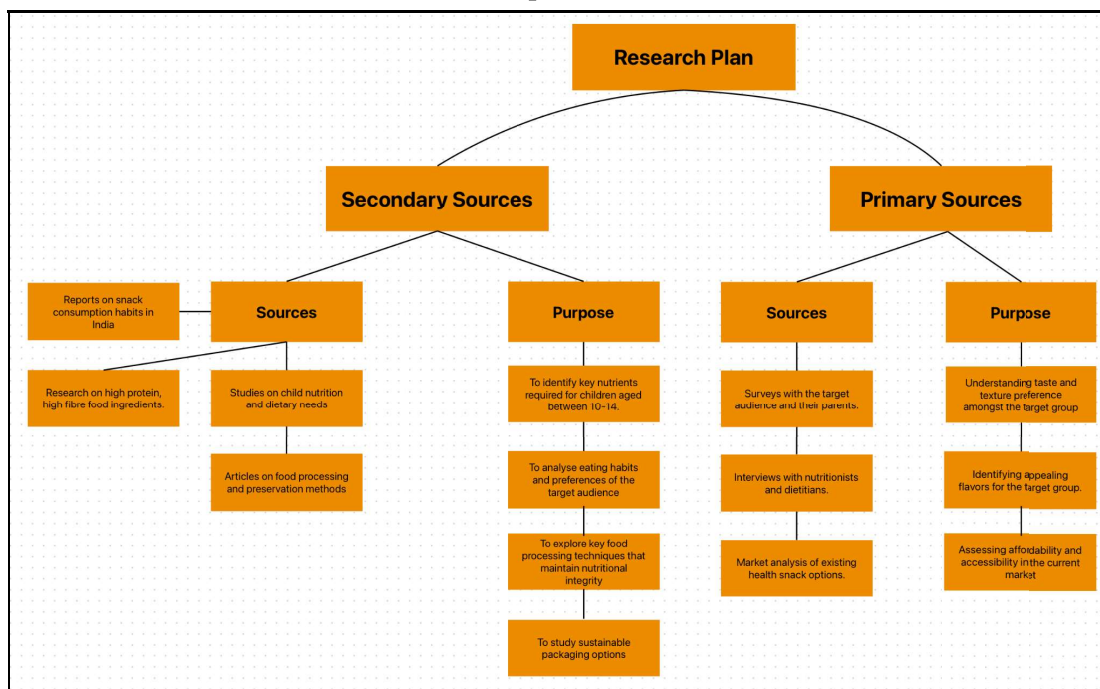
childhood diabetes cases, recording more than 10 cases per 100,000 people in urban areas. Chennai, a large metropolitan city, recorded an increase in cases, currently at 10.5 cases per 100,000 residents, peaking at ages between 10 and 12. This growing nation has also recorded a historic increase in cardiovascular disease, a study conducted in 2021 reported India had the highest number of childhood cardiovascular disease (CVD) cases, standing at 261,465, noting an increase in childhood CVD incidence by 1.28% annually. These are grave indicators of upcoming health issues that will plague Indian society if not tackled in a timely manner. *(Refer to Appendix C)*

Thus, it is necessary to create a snack, which can replace or reduce the regular consumption of ultra-processed, artificial junk by the target age group of 10 to 14 year old children in order to pave the way for a lifetime of healthy living, cutting short the deficiencies and minimising the long-term impact of malnutrition. These snacks are the most convenient mode of nutrition delivery - being easy to manufacture at large volumes, shelf-stable for months together, and most importantly, will be the path of least resistance for the target group. It is also rather convenient that the Indian market has few competitors in the healthy savoury snack category, with most market players focussing on protein bars or more traditionally 'healthy' and 'organic' produce such as unprocessed nuts, oats, cereals, etc. Even within this small group of savoury healthy snack brands, no brand markets themselves towards children. Instead they choose to attract the athletes, the gym enthusiasts or the health-conscious adults. This rare situation makes the market the most receptive it may ever be towards a 'savory snack' specially created for children aged between 10 and 14.

However, the need is not only for a savoury snack for the young, it is for a snack that is:-

- Rich in iron to reduce the risk of anemia
- Rich in fibre to reduce gut problems and ensure natural weight reduction.
- High in protein to support muscle growth and overall bodily development and function in arguably the most critical stage of a human's life.
- Low in fats to prevent excess calorie intake and reduce the risk of weight gain.
- Balanced carbohydrate quantity to provide much needed energy.
- Affordably priced, allowing for a wide range of consumers to purchase and reap the benefits.
- Flavourful, attracting the target group by providing flavours of their liking without increasing the amount of sugar or salt content.
- Attractively packaged to grab attention and encourage the young to try out this new innovation.

Strand 2: Construct a detailed research plan



Question	Priority	Source	Justification
Developing Understanding on the Topic			
How do proteins contribute to the development of adolescent children?	Medium	Website (Refer Appendix C3-12)	It is essential to understand the importance of proteins in the development of children, especially in the adolescent phase. In determining the importance of protein consumption, the value of compromising on flavour and texture in favour of protein can be determined. This would help in achieving the overarching goal of providing sustainable nutrition.
What is the daily recommended macronutrient and micronutrient intake?	Medium	Website (Refer Appendix C)	In order to design a healthy and nutrient-rich snack, the amount of nutrients one serving should contain must be determined. Understanding the recommended nutrient intake helps develop the snack in a manner compliant with regulations and industry benchmarks.
How is fibre integral to the functioning of the gastrointestinal organs?	Medium	Website (Refer Appendix C)	Fibre is touted as an important nutrient for the functioning of the digestive system, however, truly understanding the mechanism through which it aids the process is crucial to determining how much must be in one serving of the snack.
Nutrition			
What amount of nutrients (micro and macro) must be present within 1 serving of the snack?	High	Primary Ms Dhara Mavuleti, a renowned nutritionist, specialising in the needs of adolescent children. (Refer Appendix A1)	In order to design a healthy and nutrient-rich snack, a thorough understanding of the nutritional needs of the human body is required. Through understanding the needs of the human body, one can approximate the amount of nutrients (macro and micro) in one serving. This would allow the snack to provide consumers with nutrient deficiencies a chance to replenish the nutrient.
What is the maximum amount of fat that can be allowed for 1 serving?	Medium	Website (Refer Appendix C)	The amount of fat consumed in a day is paramount to the health and fitness of a person. A snack designed to help improve the dietary practices of its consumers mustn't contain unnecessary fat, and adhere to widely approved limits regulated by national governments.
What is the minimum and maximum number of calories each serving must contain to offset the nutrition from 1 meal?	Medium	Website (Refer Appendix C)	In order to remain healthy, each individual must consume a certain amount of calories and nothing more. A snack that aims at bettering the health of its consumers must adhere to standard limits set by national governments.
How can the nutritional quantity be bolstered while preventing excess calorie consumption?	High	Website (Refer Appendix C)	In order to lower calorie quantities and increase nutritional quantities, the necessary ingredients must be selected to provide that nutrition.
Preparation			
What can be the primary base of the snack itself?	High	Primary Ms Dhara Mavuleti, a renowned nutritionist, specialising in the needs	The most important part of a snack is always the base from which it is made, determining the amount of calories and nutrients per serving. By determining the most apt base, e.g. chickpea flour, the snack can

		of adolescent children. (Refer Appendix A1)	incorporate protein with ease while using supplementing agents to add the remaining nutrients.
Which fruits or vegetables can be used to provide macro and micro nutrients?	Medium	Primary Ms Dhara Mavuleti, a renowned nutritionist, specialising in the needs of adolescent children. (Refer Appendix A1)	Supplementing ingredients are incredibly important in determining the nutritional content and flavour profile of snacks. Choosing the correct ingredients can ensure cost-effective production, calorie efficient nutrition, and good flavour.
What flavours are the most appealing to children aged 10-14?	Medium	Website (Refer Appendix C)	Children aged between 10-14 tend to be picky eaters, resisting the consumption of many traditional or conventional homemade flavours. Conducting in depth research of the target audience can help determine the best flavours to develop, and thus appeal to the masses.
What cooking process would be the most effective at maintaining nutritional quantities and using the minimum fat?	High	Website (Refer Appendix C)	Through conducting in-depth research into the various cooking methods and processes that allow for the reduction of calories and most appealing flavour profile, the snack can appeal to the masses.
Packaging & Storage			
Which preservative would be the most effective at expanding shelf-life yet not harming the consumer?	Low	Website (Refer Appendix C)	The Indian market tends to reward long term shelf-stable products. Long term storage of foods gives retailers an opportunity to sell higher volume with less risk, while also allowing buyers to store the product for longer. Such added conveniences would further bolster the product's market appeal.
Is there a method of storage that would allow preservatives to be forgone?	Low	Website (Refer Appendix C)	It is well known that preservatives have been deemed unhealthy across the world, and in an ideal situation - they must be forgone. In order to further bolster the healthy image of the product, determining a form of storage which allows for it to be forgone is crucial.

Strand 3: Analyse a range of existing products that inspire a solution to the problem

Aspects for Analysis

Nutritional Profile The basic indicators of nutrition such as protein, iron, fibre, carbohydrates, fats, calories and any other key parameters..	Ingredients and Processing The type of ingredients used (artificial vs natural), the means of sourcing, extent of processing and usage of preservatives.	Taste and Texture Feedback from longtime consumers to reflect on the flavour profile of the product, mouth feel (crispy, crunchy,etc).
Pricing and Affordability The cost of 1 unit of the product, the apparent value for money and accessibility by the target market.	Branding and Positioning The marketing strategy behind the product, the way through which the message is conveyed, and the contributing factors.	Market Reach The ability of the product to reach their target market, the people who would benefit from such products and those who need them.



Product 1: Snackible

The Indian snack company Snackible provides better snack alternatives made for modern consumer groups along with affordability and portability. The company maintains a distinct